# Venkat Sai Subash Panchakarla

Cincinnati, OH • +1-740-909-9847 • venkatsaisubash.2003@gmail.com • linkedin.com/in/venkat-sai-subash-panchakarla-b166ba23a • github.com/venkatsubash2003

## PROFESSIONAL SUMMARY

Recent M.S. graduate (GPA 4.0) specializing in low-latency LLM inference, Retrieval-Augmented Generation (RAG), and distributed vector search systems. Built production-grade AI pipelines using Llama 3.1, LangChain, PyTorch, and FAISS, cutting inference latency by 35% and significantly reducing search-to-answer latency. Experienced in fine-tuning transformer architectures, prompt engineering, MLflow experiment tracking, and deploying cloud-native ML systems on AWS. Passionate about building high-impact AI systems that bridge research and production.

## EDUCATION

**University of Cincinnati** Jan. 2025 – Apr. 2026 | Graduated

*M.S. Information Technology | GPA: 4.0 | Cincinnati, OH*

Relevant Coursework: Machine Learning & Data Mining, Deep Learning, Natural Language Processing, Cloud Computing, Azure Data Engineering, IT Research Methodologies

## TECHNICAL SKILLS

**AI/ML & Data Science:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, LoRA/PEFT, Weights & Biases, XGBoost, AutoEncoders, NLP, Computer Vision, Statistical Modeling, A/B Testing

**MLOps & Infrastructure:** RAG Pipelines, FAISS, Pinecone, Weaviate, LLM Fine-Tuning, Prompt Engineering, Transformer Architectures, LLM Evaluation, Model Quantization, Embeddings, MLflow, vLLM, TensorRT, Docker, Airflow, CI/CD, Kubernetes

**Data Engineering:** SQL, Python, Spark, dbt, Snowflake, BigQuery, PostgreSQL, AWS (EC2, S3, Athena), Azure Synapse, ETL/ELT Pipelines, Feature Stores

**Backend & APIs:** FastAPI, REST APIs, OpenAI API, Asynchronous Services, Bash, C++, JavaScript, Git, Linux/Ubuntu, Jupyter Notebook

## EXPERIENCE

**ML Engineer / Data Analyst** Mar. 2025 – Present

*University of Cincinnati | Cincinnati, OH*

- Built scalable ML-ready data pipelines using Snowflake, BigQuery, and cloud-native ETL workflows, reducing manual preprocessing effort by 30% and accelerating model training cycles.
- Developed automated schema validation and anomaly detection systems (Scikit-learn, Python), cutting upstream data inconsistencies by 25% and improving downstream model reliability.
- Engineered centralized feature stores and reusable feature engineering workflows, reducing ML experiment setup time by approximately 40% and enabling faster, reproducible model iteration.
- Optimized large-scale ingestion and transformation pipelines for distributed analytical workloads, improving pipeline throughput by approximately 20% and reducing end-to-end data latency.
- Collaborated cross-functionally using Grafana and Prometheus to enhance ML pipeline monitoring and observability, reducing mean time to detect pipeline failures by 35%.

**AI/ML Engineer Intern** Apr. 2023 – Apr. 2024

*Cybereconn | Gurgaon, India*

- Designed and optimized low-latency inference pipelines for Llama 3.1 and BERT-based models, improving runtime scheduling and execution graphs, reducing end-to-end latency by 35%.
- Implemented semantic chunking and ranked retrieval architectures using BERT embeddings for AI-powered search, improving contextual relevance scores by 40% via systematic LLM evaluation.
- Developed scalable FastAPI backend services for real-time LLM inference and retrieval; engineered memory-efficient embedding workflows, improving production throughput by approximately 25%.
- Built automated evaluation and feedback pipelines using real-world interaction metrics, increasing retrieval accuracy and system reliability by 30%.
- Conducted latency profiling and performance benchmarking across distributed AI workloads, identifying runtime bottlenecks and reducing P95 inference latency by approximately 20%.
- Engineered end-to-end Retrieval-Augmented Generation (RAG) pipelines integrating FAISS vector indexing with Llama 3.1, supporting 10x larger knowledge bases and maintaining stable performance under high query concurrency.

## PROJECTS

**CareerPilot AI** *(LangChain · Ollama Llama 3.1 · Streamlit · Python · SQLite):* Built AI-powered job application assistant for international students, integrating LLM-driven visa sponsorship classification, H1B history lookup, and ATS score calculation, reducing manual job research effort by approximately 70%; features resume-JD fit analysis, cover letter generation, and recruiter email drafting backed by SQLite application tracker.

**LLM-Based Research Paper Analysis Engine** *(LangChain · FastAPI · Python · NLP):* Developed AI platform for summarizing, clustering, and comparing research papers using LLM-driven semantic analysis, cutting manual review effort by 75% and improving insight extraction quality by 45%.

**Crime Similarity & Risk Prediction System** *(AutoEncoders · Scikit-learn · Python):* Built unsupervised deep learning system using AutoEncoders across 2M+ records; optimized feature engineering pipelines (normalization, imputation, encoding) improving pattern detection accuracy by approximately 35% and enabling real-time risk scoring.

**Farmers Market Analytics & Dashboard System** *(Tableau · BigQuery · SQL · Smartsheets · Python · Scikit-learn):* Developed AI-assisted Tableau dashboards and scalable BigQuery ETL pipelines consolidating Smartsheets forms into unified datasets, improving stakeholder reporting efficiency by approximately 45%; automated KPI tracking for vendor engagement, attendee demographics, and benefits utilization, reducing manual reporting effort by approximately 60% and improving cross-dataset accuracy by approximately 30%.